A c_0-Sum Extension for the Real Lipschitz Numerical Index Question

Automatic math-discovery packet

June 12, 2026

Source question

The source paper is Antonio Pérez-Hernández, *Lipschitz vs Linear Numerical Index in certain Banach spaces*, arXiv:2507.18208. The paper studies whether the Lipschitz numerical index and the classical numerical index coincide.

$$\max_{|\theta|=1} \| \operatorname{Id}_X + \theta T \| \geq 1 + \alpha \|T\|$$

for every operator T, see e.g. [17, Proposition 3.3]. In particular, if $n(X) > 0$ then Id_X is a strongly extreme point of the unit ball of $\mathcal{L}(X)$.

These notions have been extended in several directions. In this work, we focus on a non-linear variant introduced by X. Huang, D. Tan, and R. Wang in [20], known as the *Lipschitz numerical index.* Let $\operatorname{Lip}_0(X)$ denote the Banach space of all Lipschitz maps $F : X \longrightarrow X$ such that $F(0) = 0$, equipped with the norm

$$\|F\|_L := \sup\left\{ \frac{\|F(x_1) - F(x_2)\|}{\|x_1 - x_2\|} : x_1, x_2 \in X, x_1 \neq x_2 \right\} < \infty.$$

For every $F \in \operatorname{Lip}_0(X)$, the *numerical range* of F is defined by

$$W_L(F) = \left\{ \frac{\langle x^*, F(x_1) - F(x_2)\rangle}{\|x_1 - x_2\|} : \; x^* \in S_{X^*}\,,\; x_1 \neq x_2 \in X\,, \right.$$
$$\left. \frac{\langle x^*, x_1 - x_2\rangle}{\|x_1 - x_2\|} = 1 \right\},$$

and its *numerical radius* by

$$w_L(F) = \sup\{|\lambda| \colon \lambda \in W_L(F)\}.$$

The *Lipschitz numerical index* of the Banach space X is then defined by

$$\begin{aligned} n_L(X) &= \inf\{w_L(F) \colon F \in \operatorname{Lip}_0(X), \|F\|_L = 1\} \\ &= \max\{\lambda \geq 0 \colon \lambda\|T\|_L \leq w_L(T) \text{ for every } T \in \operatorname{Lip}_0(X)\}. \end{aligned}$$

Notice that the canonical inclusion $\mathcal{L}(X) \hookrightarrow \operatorname{Lip}_0(X)$ is a linear isometry, and that if F is linear $W_L(F) = W(F)$ and $w_L(F) = w_L(F)$. Therefore,

$$n_L(X) \leq n(X)\,.$$

A central question posed in [20] is whether equality always holds. The answer to this question depends on whether the underlying field is $\mathbb{R}$ or $\mathbb{C}$.

Figure 1: The paper defines the Lipschitz numerical index and records the central equality question.

index of $(\mathbb{C}^2, \|\cdot\|_\infty)$ is one (by the aforementioned result of Kadets et al.), while the Lipschitz numerical index of $(\mathbb{C}^2, \|\cdot\|_2)$ is zero, as shown above. It is therefore natural to ask which values are realized by the pair $(n(X), n_L(X))$ as X ranges over all complex Banach spaces. However, we do not pursue this question in the present work.

Real case. In contrasts to the complex case, no known example shows that $n_L(X)$ and $n(X)$ may be different when X is a real space. This motivates the conjecture $n_L(X) = n(X)$ for every real Banach space X. In fact, partial results support this claim: Wang et al. proved that if X has the Radon-Nikodým property (RNP) then the equality holds; moreover, they showed that if X is a real lush space, then both indices coincide and are equal to one. In particular, if X does not contain a copy of ℓ_1 or X is *strongly regular*,

Figure 2: The real case is stated as the conjectural equality $n_L(X) = n(X)$ for every real Banach space X.

In this paper, we contribute further evidence to this conjecture. We prove that the equality $n(X) = n_L(X)$ holds if X is a (real) separable Banach space (Section 2) or if X is a (real) dual Banach space (Section 3). The arguments are based on two distinct techniques for constructing linear operators from Lipschitz maps that preserve certain properties: differentiation via convolution with a Gaussian probability measure, and the use of invariant means.

2. Real separable Banach spaces

We first state the main result of the section.

Figure 3: The source paper's Theorem 2.1 proves the equality for real separable Banach spaces.

Claim

Theorem 1. *Let $\{X_\gamma : \gamma \in \Gamma\}$ be a family of separable real Banach spaces and let*

$$X = \left(\bigoplus_{\gamma \in \Gamma} X_\gamma \right)_{c_0}.$$

Then

$$n_L(X) = n(X).$$

This is a partial result toward the real-case question in the source paper. It may apply to nonseparable and nondual spaces, for instance when Γ is uncountable. It does not settle the conjecture for arbitrary real Banach spaces.

Inputs used

We use two external facts.

1. The source paper proves that $n_L(Y) = n(Y)$ for every real separable Banach space Y.
2. The classical numerical index is stable under c_0-sums in the form

$$n\left(\left(\bigoplus_{\gamma \in \Gamma} X_\gamma \right)_{c_0} \right) = \inf_{\gamma \in \Gamma} n(X_\gamma).$$

The second item is the main structural input to check independently during human review.

Proof

It is always true that $n_L(X) \le n(X)$, because each bounded linear operator is a Lipschitz map fixing the origin and the two numerical radii agree on linear maps. We prove the reverse inequality.

Let $F \in \mathrm{Lip}_0(X)$ satisfy $\|F\|_L = 1$, and fix $\delta > 0$. Choose $x, y \in X$, $x \neq y$, such that

$$\frac{\|F(x) - F(y)\|}{\|x - y\|} > 1 - \delta.$$

For $z \in X$, write $z = (z_\gamma)_{\gamma \in \Gamma}$. Since X is a c_0-sum, the support

$$\mathrm{supp}(z) = \{\gamma \in \Gamma : z_\gamma \neq 0\}$$

is countable: it is the union over $m \ge 1$ of the finite sets

$$\{\gamma \in \Gamma : \|z_\gamma\| \ge 1/m\}.$$

Let

$$A = \mathrm{supp}(x) \cup \mathrm{supp}(y) \cup \mathrm{supp}(F(x)) \cup \mathrm{supp}(F(y)).$$

Then A is countable. Put

$$Y = X_A = \left(\bigoplus_{\gamma \in A} X_\gamma \right)_{c_0}.$$

The space Y is separable, because it is a countable c_0-sum of separable spaces. Let $P_A : X \to Y$ be the coordinate projection, $(P_A z)_\gamma = z_\gamma$ for $\gamma \in A$. This projection has norm one, and P_A fixes x, y, $F(x)$, and $F(y)$.

Define
$$G : Y \longrightarrow Y, \qquad G(u) = P_A F(u).$$
Then $G(0) = 0$, $\|G\|_L \leq \|F\|_L = 1$, and the chosen pair gives
$$\|G\|_L \geq \frac{\|G(x) - G(y)\|}{\|x - y\|} = \frac{\|F(x) - F(y)\|}{\|x - y\|} > 1 - \delta.$$

We next compare the Lipschitz numerical radii. Take $u, v \in Y$, $u \neq v$, and $u^* \in S_{Y^*}$ such that $u^*(u-v) = \|u-v\|$. Set $x^* = u^* \circ P_A$. Since $P_A|_Y = \mathrm{Id}_Y$ and $\|P_A\| = 1$, we have $x^* \in S_{X^*}$, and
$$x^*(u - v) = u^*(u - v) = \|u - v\|.$$
Moreover,
$$u^*(G(u) - G(v)) = u^* P_A(F(u) - F(v)) = x^*(F(u) - F(v)).$$
Thus every admissible value appearing in the Lipschitz numerical range of G is also an admissible value for F. Hence
$$w_L(G) \leq w_L(F).$$

By the source paper's separable theorem, $n_L(Y) = n(Y)$. Therefore
$$w_L(F) \geq w_L(G) \geq n_L(Y)\|G\|_L = n(Y)\|G\|_L.$$
Using the c_0-sum formula for the classical numerical index,
$$n(Y) = \inf_{\gamma \in A} n(X_\gamma) \geq \inf_{\gamma \in \Gamma} n(X_\gamma) = n(X).$$
Consequently,
$$w_L(F) \geq n(X)(1 - \delta).$$
Letting $\delta \downarrow 0$, we obtain $w_L(F) \geq n(X)$. Since F was an arbitrary norm-one element of $\mathrm{Lip}_0(X)$, this gives $n_L(X) \geq n(X)$. Together with the general inequality $n_L(X) \leq n(X)$, the equality follows.

Human review checklist

- Confirm the standard formula for the classical numerical index of c_0-sums.
- Check that the support-countability argument is valid for arbitrary c_0-sums of Banach spaces.
- Check the comparison $w_L(G) \leq w_L(F)$, especially the extension of $u^* \in Y^*$ to $u^* \circ P_A \in X^*$.
- Confirm that the result is genuinely outside the source paper when Γ is uncountable and the resulting c_0-sum is neither separable nor a dual space.

Computational verification

No numerical verification is needed for this packet. The only code used was for source handling: downloading the arXiv PDF, extracting the relevant pages, and rendering the screenshot crops included above. The mathematical verification is the coordinate-reduction argument in the proof.